Abstract Submission Form

MCAS308: Project Phase – I

1. Project Title

AI resume critiquer

2. Group members

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3. Abstract

This project develops an " AI Resume Analyzer " using modern Large Language Models (LLMs) and NLP techniques. The system extracts content from PDF and plain-text resumes, parses and normalizes sections (contact, education, experience, skills, projects), and measures alignment with a given job description. It produces a suitability score, highlights strengths and gaps, detects ATS issues (formatting and missing keywords), and generates tailored recommendations and phrasing improvements. A user-friendly Streamlit web app allows job seekers and recruiters to upload resumes and job descriptions to receive instant feedback. The tool accelerates hiring decisions, reduces bias, and guides candidates to optimize resumes for specific roles and ATS systems.

4. Problem Definition

Manual resume screening is slow and often biased, making it difficult for recruiters to efficiently identify suitable candidates. Job seekers also face challenges in optimizing resumes for specific job roles and ATS systems. An automated solution is needed to evaluate resumes, score their relevance, and provide improvement suggestions.

5. Requirements

Requirement	Details
Equipment/Tools	Processor: Intel i5/i7 or equivalent RAM: Minimum 8 GB (16 GB recommended for deep learning models like LSTM) Storage: Minimum 20 GB free space

	Graphics Card: Optional, but GPU (NVIDIA) recommended for faster neural network training
Data/Resources	Yahoo Finance (via yFinance Python library)
Software Requirements Programming Language IDE/Notebook Libraries:	Operating System: Windows / Linux / macOS Python 3.x Jupyter Notebook / PyCharm / VS Code streamlit, PyPDF2, io, os, openai, dotenv

7. Applications

1. Automatically extract structured data from uploaded resumes (contact, education, experience, skills, Certifications).

2. Compute a numeric suitability score comparing resume → job description (0–100).

3. Generate specific, editable suggestions: wording, structure, keyword inclusion, ATS changes, and a one-paragraph rewritten summary/summary bullets.

4. Make results explainable: show which sections lowered/raised the score and why.

5. Evaluate system accuracy against human reviewers and measure improvement in ATS pass rate and recruiter time saved.

8. Expected Outcomes

English-language resumes (PDF, DOCX, TXT).

Job descriptions in English.

Provide scores, suggestions, and short rewritten text snippets.

9. Guide Details

Guide Name	Contact Number	Email ID
Prof VJ kadam	8408855351	vjkadam@dbatu.ac.in

10. Submission Details

Date of Submission	[Enter the date]
27 th November	2025

Declaration

We hereby declare that the project topic and abstract submitted are original and have not been copied or plagiarized. Both group members will actively participate and adhere to the guidelines provided by the faculty.

Signatures of Group Members:

1. _____

2. _____